

Assignment Sheet

Task 1 – Data dependency graphs

(6+2 Points)

a) Create a data dependency graph for the following sequence of instructions:

```
I1: add t0, t1, t2
I2: mul t2, t0, t1
I3: mulhsu t3, t0, t1
I4: seqz t4, t3
I5: xori t4, t4, -1
I6: add t4, t4, t0
```

Dependency graph:

b) How many execution phases are needed at least? For this task, you may assume that there are always enough free functional units.

Your answer:

Task 2 – Circular Dependencies

(4 Points)

For instructions I_m and I_n with $m < n$ where I_m depends on the result from I_n and I_n depends on the result from I_m , meaning there is a circle in the data dependency graph, which instruction will be executed first and why?

Your answer:

Task 3 – Branch execution modes

(4 Points)

Explain the difference between branch prediction, eager execution and lazy execution in terms of branching.

Your answer:

Task 4 – Tomasulo's Algorithm

(4 Points)

How does Tomasulo's Algorithm improve efficiency in terms of resources and dependencies?

Your answer:

This assignment is due on 19.12.2025 at 10 am.